72) Suppose that f is an invertible function from Y to Z and g is an invertible function from X to Y. Show that the inverse of the composition fog is given by $(fog)^{-1} = g^{-1} o f^{-1}$.

Soln: By Defn. 10, fog is the function from X to Z. If $a \in X$, $(fog)(a) = f(g(a))$.

By Defn. 9, f is a bijection from Y to Z and g is a bijection from X to Y. By Defn. 8, f is both one-to-one and onto. g too is both one-to-one and onto. By Prob 33(a) and (b), fog is both one-to-one and onto. By Defn. 8, fog is a bijection. By Defn. 9, fog is invertible.

Let, $(fog)(x) = Z$ where $x \in X$ and $Z \in Z$.

$\therefore (fog)^{-1}(Z) = x$ (By Defn. 9)

$f(g(x)) = Z$. Let, $g(x) = y$ where $y \in Y$

$f(y) = Z$. By Defn. 9, $f^{-1}(Z) = y \Rightarrow f^{-1}(Z) = g(x)$

By Defn. 9, $g^{-1}(f^{-1}(Z)) = x \Rightarrow (fog)^{-1}(Z) = g^{-1}(f^{-1}(Z))$

$\therefore x, y, Z$ are chosen arbitrarily, $\boxed{(fog)^{-1} = g^{-1} o f^{-1}}$ □

73) Let S be a subset of a universal set U. The characteristic function fs of S is the function from U to the set $\{0,1\}$ such that $f_S(x) = 1$ if x belongs to S and $f_S(x) = 0$ if x does not belong to S. Let A and B be sets. Show that for all $x \in U$.

a) $f_{A \cap B}(x) = f_A(x) \cdot f_B(x)$

Soln: $f_{A \cap B}(x) = \begin{cases} 1 & \text{if } x \in (A \cap B) \\ 0 & \text{if } x \notin (A \cap B) \end{cases}$

Let, $f_{A \cap B}(x) = 1$. $\therefore x \in (A \cap B)$. By Defn. 2 from Sec-2.2, $x \in A \wedge x \in B$. By Defn. of AND, $x \in A$ and $x \in B$. By Defn. of characteristic function, $f_A(x) = f_B(x) = 1$. $\therefore f_A(x) \cdot f_B(x) = 1 \cdot 1 = 1 = f_{A \cap B}(x)$.

Let, $f_{A \cap B}(x) = 0$. $\therefore x \notin A \cap B$. By Defn. of not belongs to, $\neg(x \in A \cap B)$. By Defn. of negation, $\neg(\neg(x \in A \cap B))$ is False. By Double-negation Law of proposition $x \in (A \cap B)$ is False. By Defn. of AND, $x \in A \wedge x \in B$ is False.